

10(a)(i)	photoelectric effect	B1
10(a)(ii)	electron diffraction	B1
10(b)(i)	$\lambda = h / p$	M1
	h is the Planck constant	A1
10(b)(ii)	de Broglie (wavelength)	B1
10(c)(i)	$\frac{1}{2}mv^2 = eV$	C1
	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = 1.60 \times 10^{-19} \times 4800$ <u>so</u> $v = 4.1 \times 10^7 \text{ m s}^{-1}$	A1
10(c)(ii)	$\lambda = h / mv$ $= 6.63 \times 10^{-34} / (9.11 \times 10^{-31} \times 4.1 \times 10^7)$	C1
	$= 1.8 \times 10^{-11} \text{ m}$	A1